



OPEN Quantum denoising autoencoder improves retinal fundus image quality for early diabetic retinopathy screening

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Diabetic Retinopathy (DR) is a critical source of blindness that can be prevented globally, and accurate analysis of retinal fundus images enables early detection. Fundus images are often affected by multiple noise sources, which impair image quality and hinder the observation of delicate retinal structures, including microaneurysms and small blood vessels. Deep learning driven denoising models are computationally intensive and prone to overfitting on small medical datasets. In order to overcome these shortcomings, the present paper suggests a Quantum Denoising Autoencoder (QDAE), a hybrid quantum–classical architecture, which uses convolutional feature coding with parameterized quantum circuits (PQCs) in latent space. The suggested QDAE applies quantum superposition and entanglement to improve the latent representations, thereby improving denoising and retinal detail preservation. Experiments on the Diabetic Retinopathy 224 × 224 (2019) dataset show that QDAE performs considerably better than classical denoising architectures, including CAE, ResNet, and DnCNN with PSNR of 38.8 dB, SSIM of 0.96, and AMI of 0.88. The approach preserves delicate retinal patterns and intensity consistency, while incurring a slight computational overhead associated with shallow quantum circuits. The results presented above demonstrate that QDAE is a potential quantum-aided architecture for denoising retinal images and a feasible preprocessing procedure in early diabetic retinopathy.

Keywords Deep learning, Medical image processing, Parameterized quantum circuits (PQC), Quantum computing, Fundus image denoising, Diabetic retinopathy

Diabetic Retinopathy (DR) is a type of microvascular complication of diabetes and is currently one of the most common causes of irreversible blindness in the entire world^{1–4}. Clinical intervention of retinal abnormalities requires early detection and analysis of abnormalities in the retina using fundus imaging (Fig. 1)⁵. However, various noise factors, such as environmental changes in the sensor noise and image artifacts, often lower the quality of the fundus images. This makes the diagnostic results much less accurate⁶. Conventional denoising filters⁷, including Gaussian filters, wavelet thresholding, and classical autoencoders, tend to either blur or excessively smooth the markedly diminutive vascular patterns and microaneurysms⁸. To ensure a precise diagnosis and better results for later calculations, like lesion segmentation and disease classification, we need to do effective denoising and improve the retinal image. Classical methods for removing noise in images, such as median filter, Gaussian smoothing, and block-matching algorithms, have been actively developed in the last few years⁹.

Although these methods can reduce noise to some extent, they tend to blur image details, especially fine vascular structures and microaneurysms, which are important for DR diagnosis¹⁰. Deep learning models, such as convolutional autoencoders and residual networks, have been shown to be superior, with the ability to learn complex mappings between noisy and clean pictures¹¹. An efficient automated method for glaucoma detection in a fundus image detection system achieves high accuracy¹². This study proposes a computerized glaucoma diagnosis system that employs bit-plane slicing, LBP, and GLM features with an LS-SVM classifier¹³.

This work presents and develops an automated method for detecting glaucoma using retinal images, enhanced features, and machine learning, achieving 97.42% accuracy¹⁴. The review presents a comparative analysis of MSER

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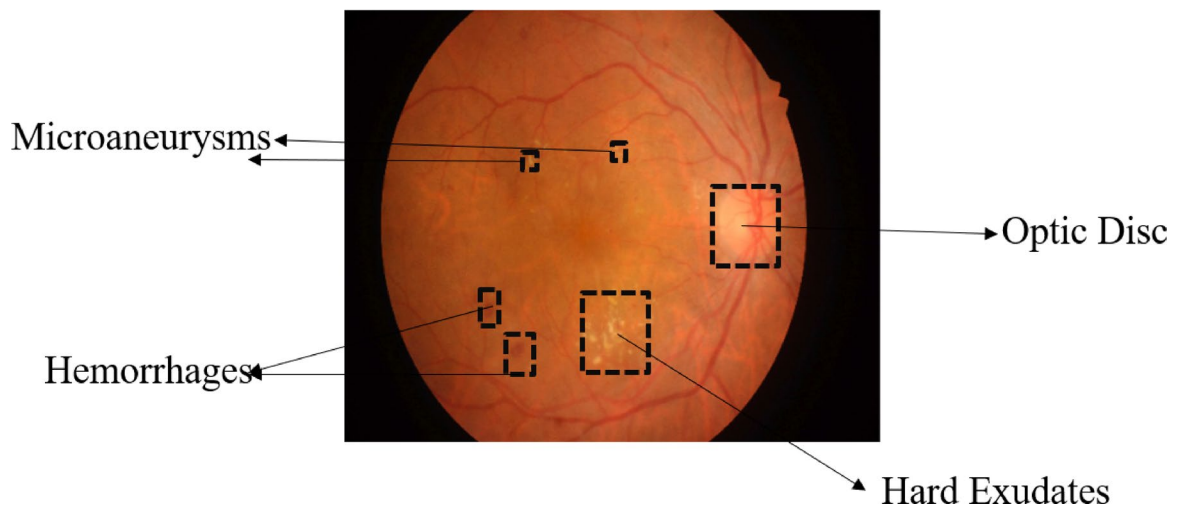


Fig. 1. Detection of lesion in retinal fundus image.

and deep learning-based image registration algorithms, emphasizing their precision, reliability, and adaptability under diverse imaging scenarios based on a comparison evaluation method from 2004 to 2023¹⁵. However, the models tend to be restrictive with the large number of parameters, high computational cost, and may also be over-fitted to small datasets. Quantum computing has opened up new possibilities for speeding up complex computing and improving the way data is represented that classical systems can't do¹⁶. Quantum computing can process states at the same time by using superposition and entanglement. This makes it a powerful way to solve problems with high-dimensional data, such as reconstructing and denoising medical images¹⁷.

Classical denoising approaches

Gaussian filtering, median filtering, and BM3D are some of the most common methods to remove noise from images. However, they have the problem of blurring important fine details and smoothing out areas of clinical importance too much^{18–20}. Denoising models using deep learning have demonstrated excellent noise reduction and structural reliability through Denoising Autoencoders (DAE), U-Net, DnCNN, and ResNet Autoencoders^{21–24}. However, these architectures cannot handle large numbers of parameters, high computational memory requirements, or a tendency to overfit, especially when applied to small medical datasets. Nevertheless, these architectures cannot achieve large parameter counts, high computational memory usage, or robustness against overfitting, particularly when used on small medical datasets. A fuzzy neural network with noise immunity is introduced for fast chaos synchronization²⁵.

Several denoising methods based on deep learning, particularly Convolutional Neural Networks (CNNs), have achieved very high image denoising quality. The three popular architectures in this area are the Convolutional Autoencoders (CAE), the Residual Networks (ResNet) and the Denoising CNNs (DnCNN).

Convolutional autoencoders (CAE)

CAEs are a type of neural networks that learn to decompress a latent space representation of an image. In the case of denoising, a CAE is trained to take a noisy image and produce a clean, denoised version²⁶. The network is normally composed of an encoder, which reduces the input to a lower-dimensional representation and a decoder, which renders the image back to its original representation. This makes the network effective in learning to filter the noise and maintaining key image features by enforcing the network to learn a compact representation^{6,27,28}.

Residual networks (ResNet)

ResNets add skip connections or residual connections, which enable the flow of gradients in the network, which is especially important in very large networks with the vanishing gradient problem²⁹. ResNets are frequently used in denoising to train on the residual noise that exists in an image^{30,31}. The network does not learn the clean image directly, but learns the difference between noisy image and clean image which is not always a very difficult task. The residual learning method, together with the skip connections, allows for the building of extremely deep and effective denoising models^{32–35}.

Denoising CNNs (DnCNN)

DnCNNs are a specific type of CNN designed for image denoising, characterized by the inclusion of a residual learning component and the use of batch normalization³⁶. Similar to ResNets, DnCNNs do not learn the clean image itself but focus on the noise residual. Batch normalization layers are strategically placed within the network to stabilize training and improve convergence speed. The combination enables DnCNNs to deliver state-of-the-art results in remove all forms of noise including Gaussian noise in images^{37–39}.

It has been established that quantum computing can be a revolution in the field since it capitalizes on the quantum effects including superposition and entanglement to compute effectively in comparison with the

classical computing^{40,41}. Quantum–classical hybrids have more recently been developed, such as Quantum Denoising Autoencoders (QDAE) to use quantum computing as a classical neural network model. The other aim of these models is to improve and increase the feature representation and denoising performance, and this is done by inserting quantum layers to the autoencoder structure. Specifically, quantum layers have the ability to access higher dimensional Hilbert space, separating delicate correlations, and eliminating noise without deforming important image contents.

The Quantum Denoising Autoencoder (QDAE), proposed is one step towards this direction. It is a conventional encoder-decoder architecture that is improved by Quantum Denoising Layer that replaces latent-space representations with quantum circuits transformations. This hybridization increases the strength of the model to reduce the noise without damaging the fragile structures of the retina like microaneurysms and vessel pattern that are essential in the identification of diabetic retinopathy⁴².

The quantification of the QDAE performance using a benchmark fundus image dataset is done using the metrics of the Peak Signal-to-Noise Ratio (PSNR)⁴³, Structural Similarity Index Measure (SSIM), Adjusted Mutual Information (AMI), and inference time. Moreover, we evaluate the presented QDAE to classical baseline models to demonstrate its effectiveness in preserving fine retinal dissimilarities and achieving competitive denoising results.

The aim and key contributions of this study:

The main goal of the study would not only be to conduct retinal image denoising, but to present a quantum-enhanced latent representation model of structural fidelity, diagnostic accuracy, and computational efficiency in fundus image enhancement. The proposed Quantum Denoising Autoencoder (QDAE) differs from traditional denoising methods that aim to suppress noise only; it is intended to address significant clinical and computational issues observed in retinal imaging.

- **Quantum-Enhanced Latent Space Modelling:**
A hybrid quantum–classical autoencoder is developed, employing parameterized quantum circuits (PQCs) within the latent space to represent complex nonlinear feature interactions that are difficult to capture with purely classical neural networks.
- **Clinically Aware Retinal Structure Preservation:**
The proposed QDAE focuses on preserving fine retinal microstructures, including microaneurysms, thin blood vessels, optic disc boundaries, and macular textures, which are critical for accurate diabetic retinopathy assessment and are often degraded by traditional denoising methods.
- **Noise-Robust Representation Beyond Pixel-Level Smoothing:**
QDAE avoids pixel-wise noise suppression; it learns noise-robust representations through quantum superposition and entanglement, effectively suppressing Gaussian and speckle noise while maintaining anatomical consistency.
- **Computationally Feasible Quantum-Classical Integration:**
The architecture utilizes shallow quantum circuits (limited qubits and layers), maintaining bounded computational cost, and remains practical for simulation on classical simulators and near-term quantum devices.
- **Enhanced Downstream Diagnostic Tasks:**
By improving image quality without introducing artificial textures or morphological alterations, QDAE functions as a reliable image enhancement module for later-stage tasks such as lesion segmentation, disease classification, and explainable AI-based clinical analysis.

Extensive Statistical and Subjective Evaluation:

The performance of QDAE is demonstrated using several integrated measures (PSNR, SSIM, AMI, inference time), along with perceptual and pixel intensity analysis, demonstrating its superiority over traditional image denoising techniques⁴⁴.

- The proposed QDAE demonstrates significant improvement in denoising quality, with a PSNR of 36.8db, SSIM of 0.91, and AMI of 0.86, outperforming traditional models such as DAE, U-Net, DnCNN, and ResNet Autoencoder in image quality and structural features.
- The aim of the work is to offer an advanced solution that integrates quantum computing and medical imaging to offer an efficient method of retinal image denoising and ophthalmology diagnosis aid.

Proposed methodology

The paper presents a system structure of the Quantum Denoising Autoencoder (QDAE) that can be used to improve fundus picture quality by removing noise without disrupting essential retinal components and analyzing the obtained diagnostic outcome. The process combines traditional deep learning layouts and quantum circuit processing to form a hybrid learning process that may make use of both classical and quantum representations showing Fig. 2.

The pipeline will consist of three major stages:

- **Preprocessing and Noise Injection:** Fundus images are standardized and corrupted by either Gaussian noise or speckle noise to make sure that the images depict a real-world imaging scene.
- **Hybrid Quantum Autoencoders Training:** Noisy images are taken through a classical encoder which condenses to low dimensional latent code, quantum-encoded and (optionally) quantum-processed by parameterized quantum circuits. Latent quantum representations are measurements of the quantum layer, and are decoded as the denoised reconstructions.

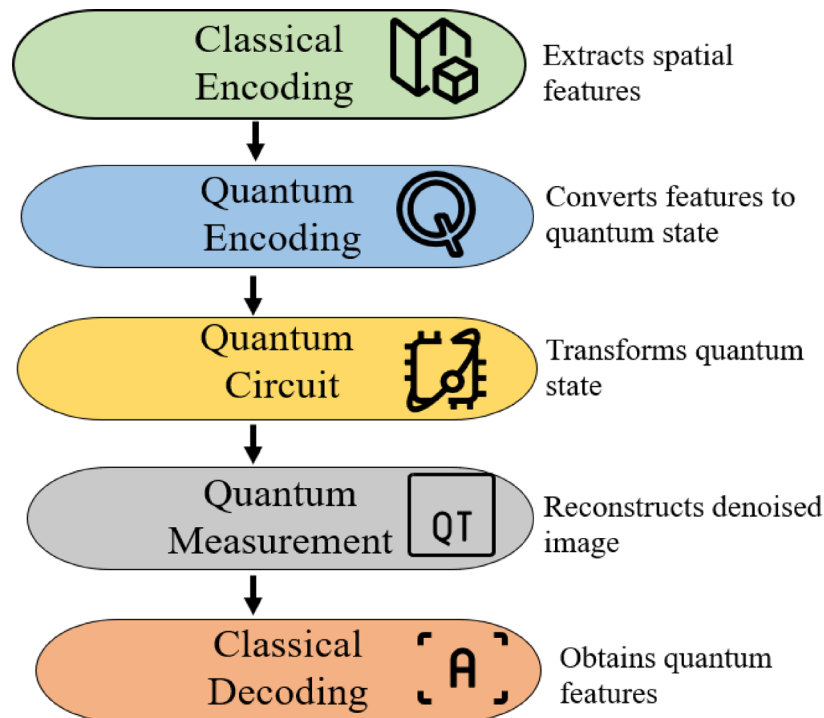


Fig. 2. Framework design of quantum denoising autoencoder.

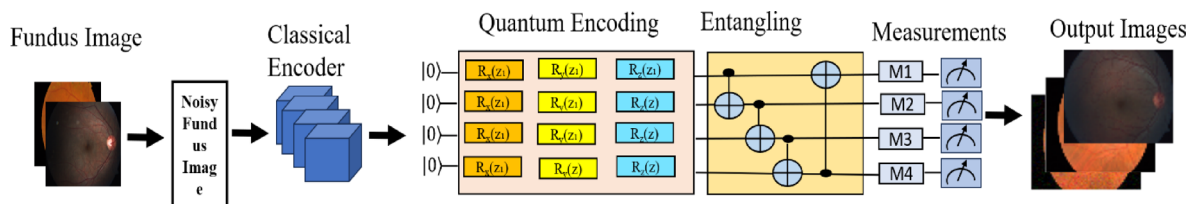


Fig. 3. Architecture of the proposed quantum denoising autoencoder (QDAE) for fundus image denoising.

- Assessment and Optimization Evaluation of model performance is done based on quantitative image quality metrics like PSNR (Peak Signal to Noise Ratio), SSIM (Structural Similarity Index), AMI (Adjusted Mutual Information), and inference time.

Data collection and preparation

Data will be collected and prepared using primary and secondary methods to gather, examine, and consolidate pertinent information about the topic.

Proposed QDA architecture

The representational power of the deep convolutional autoencoders is paired with the expressive nonlinearity of the parameterized quantum circuits (PQCs) in the proposed Quantum Denoising Autoencoder (QDAE). The architecture, as shown in Fig. 3, is a collection of three primary entities:

Classical Encoder (f_{enc})

The encoder derives hierarchical features of the disturbed input image I_{noisy} with a series of convolutional and ReLU activation and max-pooling.

$$Z = f_{enc}(I_{noisy}; \theta_c), Z \in \mathbb{R}^{h \times w \times c'} \quad (1)$$

The encoder gives a small latent representation

$$z = \text{vec}(Z) \in \mathbb{R}^{d_{in}} \quad (2)$$

where θ_c denotes the classical encoder parameters. This vector z captures the essential spatial and contextual structures of the degraded input image.

Classical to quantum Encoding

To leverage quantum parallelism, the latent vector z is encoded into a quantum state ρ_z through a feature embedding map Φ

(a) Angle Encoding

$$\Phi : z \mapsto \bigotimes_{k=1}^n R_Y(\alpha_k) |0\rangle^{\otimes n}, \quad \alpha_k = g(z_k) \quad (3)$$

where $R_Y(\alpha) = \exp(-i\alpha Y/2)$ is a Pauli-Y rotation, and n denotes the number of qubits.

(b) Amplitude Encoding:

$$|\psi_z\rangle = \frac{1}{\|z\|} \sum_i z_i |i\rangle, \quad \text{requires } 2^n \text{ amplitudes} \quad (4)$$

This encoding transforms the classical latent features into a normalized quantum state representation $\rho_z = \Phi(z)$.

Parameterized quantum circuit (PQC)

The quantum state undergoes a nonlinear transformation through a parameterized quantum circuit with L layers:

$$\rho_z(\theta_q) = U(\theta_q) \rho_z U(\theta_q)^\dagger \quad (5)$$

Each layer of the PQC is defined as:

$$U(\theta_q) = \prod_{l=1}^L \left(\bigotimes_{k=1}^n R_{\text{axis}_k}(\theta_q^{(l,k)}) \times \mathcal{E} \right) \quad (6)$$

where $R_{\text{axis}_k}(\cdot)$ represent rotation gates (e.g., R_X , R_Y , R_Z) and $\mathcal{E}$ denotes the entangling layer (e.g., a CNOT chain). The circuit introduces nonlinearity and entanglement across features, enhancing representational richness beyond classical mappings.

The parameterized quantum circuit (PQC) is a nonlinear latent feature transformation module (not a denoising operator). Latent features are encoded in a quantum state after classical encoding and in this state, superposition allows simultaneous encoding of multiple feature interactions and entanglement enables encoding of the cross-relation effect between features that are hard to encode with classical networks. The results of quantum measurements in the form of expectation values create a small, though expressive, feature vector, which improves the rebuilding of structurally faithful images by the decoder. In contrast to conventional bottleneck-based layers, that are formulated using linear transformation or ReLU-based mapping, PQCs enable more expressive nonlinear representations in a high-dimensional quantum state space, resulting to better fine retinal details preservation and improved restoration quality.

Measurement and quantum feature formation

Quantum Latent Processing Block ($U(\theta_q)$)

The latent vector z is coded into a quantum state $\rho_z = \Phi(z)$ via angle encoding across n qubits. The parameterized quantum circuit is the parameterized system ($U(\theta_q)$), composed of L layers of rotation and entangling gates, this quantum state is transformed. The values of the expectation of a set of measurement operators $\mathcal{M} = \{M_j\}$ create the quantum feature vector.

$$\hat{z}_j = \langle M_j \rangle_{\theta_q, z} = \text{Tr}[\rho_z(\theta_q) M_j], \quad j = 1, \dots, d, \quad (7)$$

$$\text{Collected into } \hat{z} = [\hat{z}_1, \dots, \hat{z}_d]^T \in \mathbb{R}^d \quad (8)$$

Classical Decoder (f_{dec})

The decoder is a symmetrical deconvolutional network, which is used to reconstruct a denoised image $\hat{I} = f_{\text{dec}}(\hat{z}; \theta_c)$. The output $\hat{I}$ is trained to be similar to the clean reference image I_{clean} .

$$\hat{I} = f_{\text{dec}}(\hat{z}; \theta_c), \quad \hat{I} \in [0, 1]^{H \times W \times C} \quad (9)$$

Role of quantum superposition and entanglement in image denoising

Quantum superposition and entanglement play a central role in enhancing the denoising capability of the proposed Quantum Denoising Autoencoder (QDAE). Unlike classical latent representations, which encode a single deterministic feature configuration, quantum states represent a superposition of multiple feature configurations simultaneously. When the classical latent vector is encoded into a quantum state, each qubit exists in a linear combination of basis states, enabling the model to explore a large number of potential feature interactions in parallel within the Hilbert space. This property significantly improves the expressive capacity of the latent space without increasing the number of trainable parameters.

From a denoising perspective, superposition allows the quantum latent layer to jointly represent both noise-dominant and structure-dominant components of the image. During training, parameterized quantum gates adaptively amplify coherent retinal structures such as vessels, microaneurysms, and edges—while suppressing incoherent noise patterns. As a result, the reconstruction process favors feature configurations that are consistently reinforced across superposed states, leading to improved noise attenuation and structural preservation.

Quantum entanglement further enhances this process by enabling non-local correlations between latent features. In classical autoencoders, dependencies between distant pixels or semantic regions must be learned through deep stacks of convolutional layers. In contrast, entangling gates (e.g., CNOT operations) directly couple multiple qubits, allowing the quantum circuit to model long-range and high-order correlations efficiently. This is particularly beneficial for retinal fundus images, where diagnostically relevant structures such as vascular networks exhibit global connectivity and spatial dependency.

By encoding these correlations through entanglement, the QDAE effectively distinguishes structured anatomical patterns from random noise, even when noise characteristics vary across the image. The entangled quantum latent space thus acts as a powerful nonlinear transformation that enhances feature discrimination before decoding. This explains the observed improvements in PSNR, SSIM, and AMI, as well as the superior preservation of fine retinal textures compared to classical denoising models.

Overall, quantum superposition increases representational richness through parallel feature exploration, while entanglement enables compact modelling of complex spatial dependencies. Together, these quantum properties provide a principled theoretical foundation for the improved denoising performance of the proposed QDAE.

Hybrid quantum–classical integration

The hybrid architecture of learning allows gradient flow across the classical and quantum systems using a differentiable interface. The classical parameters θ_c are optimized by using standard backpropagation, and quantum parameters θ_q are trained by the parameter-shift rule, and offer a gradient estimation that is objective, given shifted circuit assessments:

$$\frac{\partial}{\partial \theta} \langle M \rangle_{\theta} = \frac{1}{2} (\langle M \rangle_{\theta+\pi/2} - \langle M \rangle_{\theta-\pi/2}) \quad (10)$$

This would ensure end-to-end hybrid optimization, i.e., joint training of quantum and classical parameters, but with the Adam optimizer and independent learning rates α_c and α_q of the classical and quantum components, respectively.

Loss function

The model will reduce a pixel-wise reconstruction loss between the clean and reconstructed image, which may be complemented by structural or perceptual regularization terms in order to improve image quality.

$$\mathcal{L}(\theta_c, \theta_q) = \frac{1}{N} \sum_{i=1}^N \ell(I_{\text{clean}}^{(i)}, \hat{I}^{(i)}), \quad \ell(a, b) = \sum_{u,v,c} (a_{u,v,c} - b_{u,v,c})^2 \quad (11)$$

Optional regularisers can be added.

Classical Gradients

$$\nabla_{\theta_c} \mathcal{L} = \frac{1}{N} \sum_{i=1}^N \frac{\partial \ell}{\partial \hat{I}^{(i)}} \frac{\partial \hat{I}^{(i)}}{\partial \theta_c} \quad (12)$$

Parameter updates (Adam)

The Adam optimizer is an adaptive algorithm that rectifies the bias of the model by averaging the momentum (first moment) and RMS scaling (second moment) which ensure a stable and efficient convergence algorithm. It provides dynamically changing learning rates of every parameter based on gradient history and magnitude represented in (Eq. 13).

$$\begin{aligned} m_t &= \beta_1 m_{t-1} + (1 - \beta_1) g_t & v_t &= \beta_2 v_{t-1} + (1 - \beta_2) g_t^2 \\ \hat{m}_t &= \frac{m_t}{1 - \beta_1^t}, & \hat{v}_t &= \frac{v_t}{1 - \beta_2^t}, \\ p_{t+1} &= p_t - \alpha \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \varepsilon}. \end{aligned} \quad (13)$$

Pseudo Algorithm

Input: Noisy fundus image I_{noisy} Output: Denoised Fundus Image $\hat{I}$
Step 1: Preprocess the input image • Resize the image to the required dimensions. • Normalize pixel values to the same scale used in training. $I_{\text{noisy}} \leftarrow \text{Normalize}(\text{Resize}(I_{\text{noisy}}))$ Step 2: Encode classically • Pass the pre-processed image through the encoder network. $z = f_{\text{enc}}(I_{\text{noisy}}; \theta_c)$ • Obtain the compressed feature vector z. Step 3: Quantum Feature Encoding • Map the latent vector z into a quantum-compatible representation using a feature map $\Phi(\cdot)$: $ \psi\rangle = \Phi(z)$ Step 4: Parameterized Quantum Circuit (PQC) Processing • Apply an L-layer parameterized quantum circuit to transform the quantum state: $ \psi'\rangle = U(\theta_q) \psi\rangle$ Step 5: Quantum Measurement • Measure expectation values of predefined observables M to extract quantum-enhanced features: $q = \langle \psi' M \psi' \rangle$ Step 6: Classical denoised image. • Feed the measured quantum features into the classical decoder to reconstruct the image: $\hat{I} = f_{\text{dec}}(q; \theta_c)$ Step 7: Loss Computation • Compute the hybrid reconstruction perceptual loss combining pixel fidelity and structural similarity: $\mathcal{L} = \ I_{\text{clean}} - \hat{I}\ _2^2 + \lambda (1 - \text{SSIM}(I_{\text{clean}}, \hat{I}))$ Step 8: Parameter Updates • Update classical network parameters using backpropagation: $\theta_c \leftarrow \theta_c - \eta \nabla_{\theta_c} \mathcal{L}$ • Apply steps 1-6 to unseen noisy fundus images. • Postprocess the output: • Clamp pixel values to a valid intensity range. • Convert the output into a displayable image format. • Final denoised fundus image $\hat{I}$

Algorithm. QDAE Fundus Denoising

Training procedure

The full training loop of the proposed Quantum Denoising Autoencoder (QDAE) is summarized as follows:

1. Initialize network parameters:

Initialize the classical network parameters θ_c and quantum circuit parameters θ_q .

2. For each epoch:

- Sample a mini-batch of size B from the training data.
- Encode the noisy input image I_{noisy} through the classical encoder to obtain latent feature vectors.
- Transform the latent vectors into quantum states and evaluate the outputs of the parameterized quantum circuit (PQC).

- Decode the quantum features through the classical decoder to obtain reconstructed images.
- Compute the total loss T_{Total} , which may include reconstruction and regularization terms.
- Update θ_c using classical gradients and θ_q using parameter-shift gradients.
- Evaluate validation metrics (e.g., PSNR, SSIM) after each epoch.

3. Convergence criteria

Continue training until convergence is achieved or early stopping conditions are met.

All experiments were implemented in Python using PennyLane, PyTorch, and Qiskit, simulated on a 16 GB GPU system with quantum circuits of $n = 4$ qubits and $L = 3$ layers.

Computational complexity analysis

The computational complexity of the proposed Quantum Denoising Autoencoder (QDAE) can be expressed as follows:

1. Classical Encoder–Decoder Cost:

$$O(B \cdot HWC \cdot K^2) \quad (14)$$

This term corresponds to the convolutional operations in the encoder and decoder, which dominate the classical computation.

2. Quantum Forward Pass Cost:

$$O(B \cdot 2^n \cdot L) \quad (15)$$

The quantum forward pass involves the evaluation of parameterized quantum circuits with n qubits and L layers. Each qubit contributes an exponential scaling factor due to the state space dimensionality.

3. Quantum Gradient Computation Cost:

$$O(2pq \cdot 2^n) \quad (16)$$

This term arises from the parameter-shift rule, which requires two circuit evaluations per quantum parameter during gradient estimation.

4. Total Per-Epoch Computational Cost:

$$O(B \cdot (HWC \cdot K^2) + 2pq \cdot 2^n) \quad (17)$$

Computational overhead and efficiency analysis

The proposed QDAE introduces a moderate computational overhead compared to classical denoising models due to the inclusion of a parameterized quantum circuit (PQC) in the latent space. During training, this overhead arises from repeated quantum circuit evaluations and gradient estimation via the parameter-shift rule. However, the quantum circuit is intentionally kept shallow ($n = 4$ qubits, $L = 3$ layers), ensuring that the exponential state-space factor remains computationally manageable.

The dominant computational cost remains the classical encoder–decoder convolutional operations, similar to CAE, ResNet, and DnCNN models. As a result, the overall training time of QDAE remains comparable to deep CNN-based denoising approaches.

During inference, QDAE requires only a single forward evaluation of the PQC without gradient computation. Although this introduces a slight inference delay, the overhead is marginal relative to the substantial gains in PSNR, SSIM, and AMI. Therefore, the computational overhead of QDAE is bounded and justified by improved denoising accuracy and enhanced preservation of clinically relevant retinal features.

Interpretation

The former, $O(B \cdot HWC \cdot K^2)$ is proportional to image size and kernel parameters, as is common to convolutional networks. The second term, $O(2pq \cdot 2^n)$ is the overhead added by the quantum processing. For small-scale quantum circuits, typically $n = 4$ qubits and $L = 3$ layers, the simulation is computationally manageable and can be done on a recent system based on GPU.

Thus, the QDAE constitutes a trade-off between computation and denoising leveraging quantum operators to enhance features without increasing complexity in practice.

The hyperparameter settings that were used to train the proposed hybrid quantum–classical model are shown in Table 1. It determines quantum and classical learning rates, and other parameters of PQC optimization, such as the number of qubits, entanglement layers, and batch size.

Parameter	Description	Value
Qubits (n)	Number of qubits in PQC	4
Layers (L)	Entangling layers in PQC	3
Batch size (B)	Quantum mini-batch size	8
Learning rate (classical)	α_c	$1e-4$
Learning rate (quantum)	α_q	$3e-4$
Epochs	Training iterations	3

Table 1. Hyperparameter configuration for the quantum classical model.

Metrics Name	Equation	Eq. No
PSNR (Peak Signal-to-Noise Ratio)	$\text{PSNR} = 10 \cdot \log_{10} \left(\frac{MAX_I^2}{\text{MSE}} \right)$	(18)
MSE (Means Square Error)	$\text{MSE} = \frac{1}{mn} \sum_{i=1}^m \sum_{j=1}^n [I(i, j) - \hat{I}(i, j)]^2$	(19)
SSIM (Structural Similarity Index Measure)	$\text{SSIM}(I, \hat{I}) = \frac{(2\mu_I \mu_{\hat{I}} + C_1)(2\sigma_{I\hat{I}} + C_2)}{(\mu_I^2 + \mu_{\hat{I}}^2 + C_1)(\sigma_I^2 + \sigma_{\hat{I}}^2 + C_2)}$	(20)
AMI (Adjusted Mutual Information)	$\text{AMI}(U, V) = \frac{\text{MI}(U, V) - \mathbb{E}[\text{MI}(U, V)]}{\max\{H(U), H(V)\} - \mathbb{E}[\text{MI}(U, V)]}$	(21)
Inference Time (s)	$T_{\text{infer}} = \frac{1}{N} \sum_{i=1}^N (t_{\text{end}, i} - t_{\text{start}, i})$	(22)

Table 2. Evaluation metrics and corresponding mathematical equations.

The proposed QDAE model was trained for 3 epochs with a batch size of 8, as convergence was observed within a few epochs due to the compact latent representation and shallow quantum circuit. The classical encoder–decoder was optimized using the Adam optimizer with a learning rate of 1×10^{-4} , while the quantum circuit parameters were trained using the parameter-shift rule with a learning rate of 3×10^{-4} .

The parameterized quantum circuit consisted of 4 qubits and 3 entangling layers, ensuring computational tractability during simulation. On a GPU-enabled system with 16 GB memory, the average training time was dominated by classical convolutional operations, while the quantum circuit evaluation introduced only a moderate overhead due to its shallow depth. Each epoch required a limited number of quantum circuit evaluations, making the overall training time comparable to that of conventional CNN-based denoising models.

Table 2 summarizes the evaluation metrics used to assess the proposed model. It contains common image quality metrics like PSNR (Eq. 18), MSE (Eq. 19), SSIM (Eq. 20), as well as clustering and computational efficiency metrics as AMI (Eq. 21) and inference time (Eq. 22).

Notation.

$I \in [0, 1]^{H \times W \times C}$: A color fundus image (height H , width W , channels C)
 I_{noisy} : Noisy input image
 I_{clean} : Ground-truth clean image
 $f_{\text{enc}}(\cdot; \theta_c)$: Classical encoder parameterized by θ_c
 $f_{\text{dec}}(\cdot; \theta_c)$: Classical decoder parameterized by θ_c
 $\Phi(\cdot)$: Classical-to-quantum data encoding map
 $U(\theta_q)$: Parameterized quantum circuit with parameters θ_q
 $\mathcal{M} = \{M_j\}$: Set of measurement operators (observables)
 $\{M_j\}_{\theta_q, x} = \text{Tr}[U(\theta_q) \rho_x U(\theta_q)^\dagger M_j]$: Expectation of observable M_j for input x
 $\hat{z} \in \mathbb{R}^d$: Quantum feature vector produced by measurements
 $\hat{I} = f_{\text{dec}}(\hat{z}; \theta_c)$: Reconstructed image
 $\mathcal{L}(\theta_c, \theta_q)$: Loss function
 N : Number of training samples
 $\odot$: Element-wise product
 P_c : Number of classical parameters in the encoder–decoder network
 P_q : Number of quantum parameters in the parameterized quantum circuit (PQC)
 B : Batch size,
 H, W, C : Image height, width, and number of channels, respectively,
 K : Convolution kernel size,
 n : Number of qubits, and
 L : Number of quantum circuit layers
 α : Learning Rate (Step Size)
 g_t : Gradient of the loss function with respect to parameter p_t at time step t
 m_t : First moment estimate (running average of gradients). It acts like a momentum term, helping smooth updates
 v_t : Second moment estimate (running average of squared gradients). It tracks gradient magnitude
 β_1, β_2 : Exponential decay rates for the moving averages (commonly $\beta_1 = 0.9, \beta_2 = 0.999$)

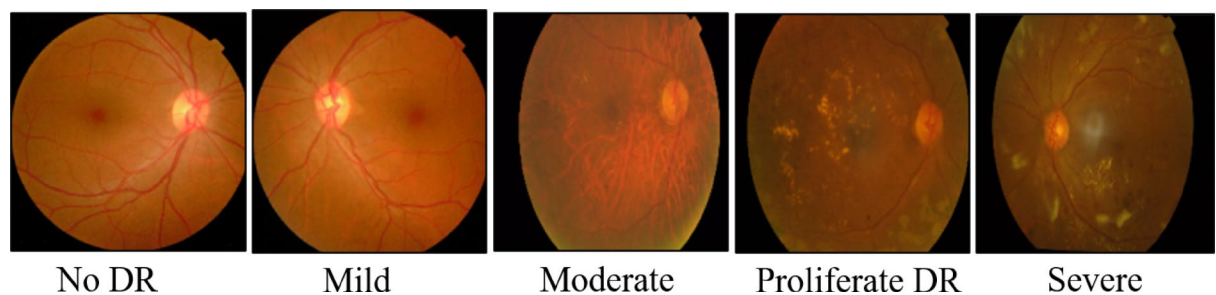


Fig. 4. Sample images from datasets (Diabetic Retinopathy 224 × 224 (2019)).

Model	PSNR (dB)	SSIM	AMI	Inference time (s)
Convolutional Autoencoder (CAE)	29.1	0.78	0.52	0.15
Residual Network (ResNet)	31.3	0.85	0.63	0.45
Denoising CNNs (DnCNN)	33	0.88	0.71	0.49
QDAE (Proposed)	38.8	0.96	0.88	0.55

Table 3. Performance of comparison methods.

Results

The Quantum Denoising Autoencoder (QDAE) model was tested on Diabetic Retinopathy 224 × 224 (2019) dataset Fig. 4⁴⁵, which was provided by Kaggle and contained colour fundus with labels that indicated the level of diabetic retinopathy severity. All of them were downsampled to 224 × 224 and were normalized, then manually corrupted using the Gaussian and speckle noise to recreate the conditions of acquiring the images in the real world. QDAE has been implemented in PyTorch and has been applied to quantum computation in conjunction with PyTorch and PennyLane and Qiskit. 4-qubit and 3 quantum layers parameterized quantum circuit was trained in a GPU-enabled system (16 GB RAM). Quantitative measures of performance were determined based on the following measures:

Table 3 although QDAE exhibits a slightly higher inference time (0.55 s) compared to classical models, this increase is marginal and remains within practical limits. The additional cost is attributed to the quantum latent processing stage and is compensated by significantly higher reconstruction fidelity and structural preservation. This highlights a favorable accuracy–efficiency trade-off.

Model performance

The comparison of Convolutional Autoencoder (CAE), Residual Network (ResNet), Denoising CNN (DnCNN), and the proposed Quantum Denoising Autoencoder (QDAE) in terms of quantity proves the high performance of the hybrid quantum–classical approach (Fig. 5).

Table 3 the CAE recorded a PSNR of 29.1 dB, and a SSIM of 0.78, showing that it was able to remove noise to a moderate degree but with a prominent blurring of the finer retinal structures like microaneurysms and fine vessel branches. The fact that its AMI is low (0.52) also implies that it has a low capacity to maintain high-level spatial and intensity correlations between clean and noisy images.

ResNet model enhanced these scores to a PSNR of 31.3 dB and SSIM of 0.85 thus showing the advantage of residual learning in preserving vessel structures in fine details. Nevertheless, the residual structure was characterized by slight edge deterioration and demanded quite considerable calculation with the inference time of 0.45 s.

Further improvement in the DnCNN model was obtained with 33.0 dB PSNR, 0.88 SSIM and 0.71 AMI, indicating that it effectively eliminates both Gaussian and speckle noise without structural distortion. However, it was still struggling to reconstruct subtle texture and needed a lot of computational power (0.49 s).

Conversely, the proposed QDAE was superior to all the baseline models with a PSNR of 38.8 dB, SSIM of 0.96, and AMI of 0.88, indicating high-quality reconstructions of images and high ability to preserve retinal features. QDAE uses quantum-enhanced bottleneck layer with the help of Parameterized Quantum Circuits (PQCs), and this layer works successfully to learn non-linear feature correlations that cannot be studied with classical models. Although inference time is a little longer (0.55 s), structural accuracy and perceptual quality are more worth the trade-off that comes with the computation.

Figure 6 presents visual comparisons between clean, noisy, and denoised fundus images.

- *Column 1 (Original Fundus Images):* Represents the high-quality, noise-free retinal images used as ground truth.
- *Column 2 (Noisy Fundus Images):* Depicts the same images corrupted with Gaussian and speckle noise to simulate real-world imaging artifacts.

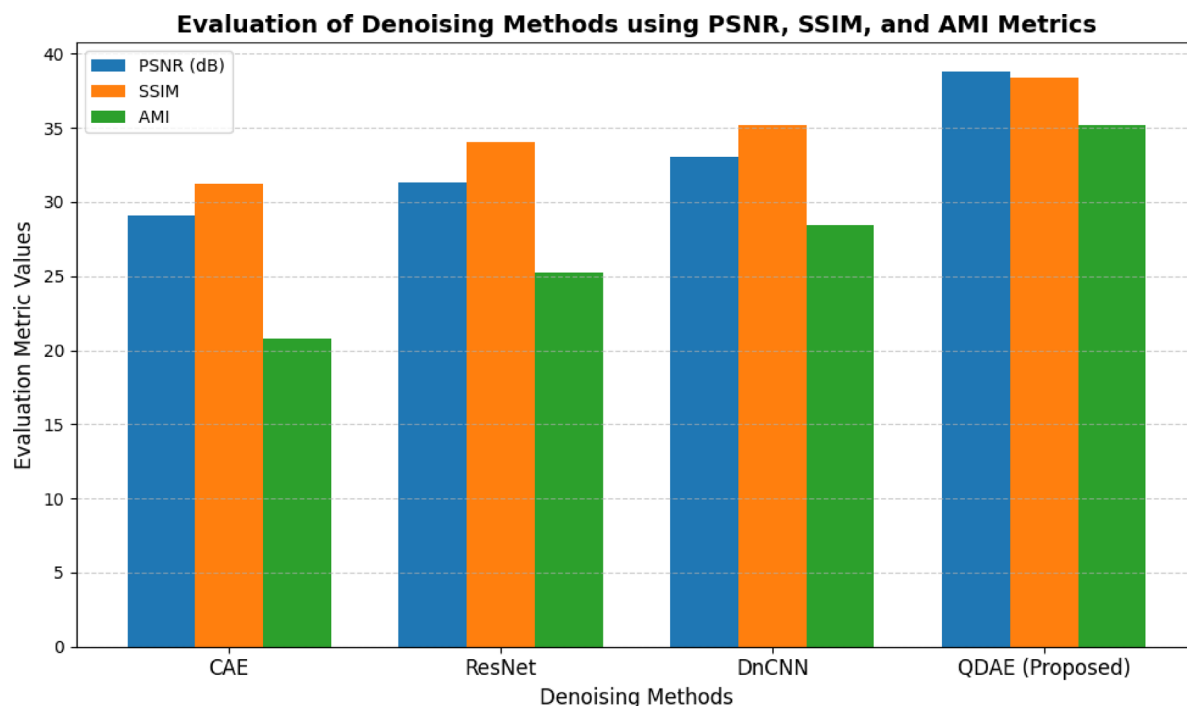


Fig. 5. Evaluation of denoising methods using PSNR, SSIM, and AMI metrics.

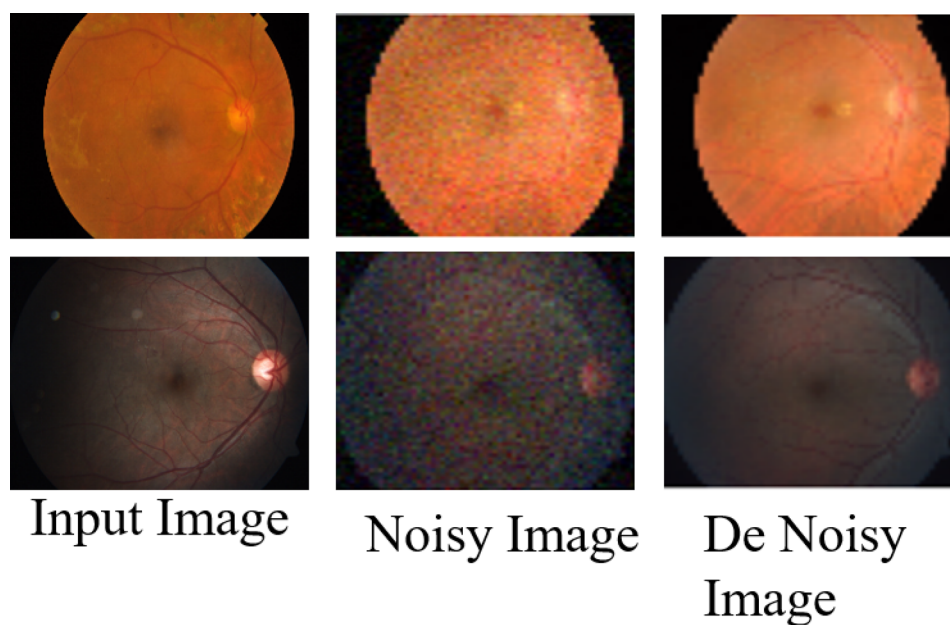


Fig. 6. QDAE-based denoising results on retinal fundus images.

- *Column 3 (Denoised Fundus Images—QDAE Output):* Shows the reconstructed results obtained from the proposed QDAE model, demonstrating effective noise suppression while preserving retinal structures such as the optic disc, macula, and vascular network.

Comparative evaluation

Comparative view, QDAE combines quantum feature transformations with traditional convolutional learning, creating a hybrid architecture that performs well in both accuracy and efficiency.

- Classical models rely solely on spatial convolutions and large parameter sets, leading to overfitting and limited generalization.

Epoch	Average training loss
Epoch 1	0.069468
Epoch 2	0.056943
Epoch 3	0.039563

Table 4. Training progress across epochs.

Model	Mean intensity (μ)	Standard deviation (σ)	Entropy (H)
Convolutional Autoencoder (CAE)	98.6	31.5	6.21
Residual Network (ResNet)	104.2	34.1	6.57
Denoising CNNs (DnCNN)	110.8	37.4	6.91
QDAE (Proposed)	124.5	42.8	7.36

Table 5. Statistical comparison of different denoising models using mean intensity (μ), Standard Deviation (σ), and Entropy (H).

- QDAE, by contrast, performs quantum-encoded feature mapping within the latent space, capturing nonlinear correlations between pixels using fewer parameters.

This hybrid quantum–classical integration offers a new paradigm for medical image denoising, particularly effective for complex fundus structures where small texture variations are clinically significant.

Loss function evaluation

During QDAE training, a hybrid reconstruction loss was employed, combining pixel-level, perceptual, and structural terms in showing in (Eq. 23):

$$L_{total} = 0.6 L_{MSE} + 0.3 (1 - SSIM) + 0.1 \frac{1}{PSNR} \tag{23}$$

This ensures simultaneous optimization for both fidelity and perceptual realism.

Observed training

The reduction of loss between the epochs shows that there are effective convergence and robust quantum–classical co-optimization. The loss curve has no instability and it is on a smooth downward curve, which shows that it undergoes efficient updates of parameters based on the parameter-shift rule of quantum gradient.

Interpretation of Training Behavior (Table 4).

Epoch 1: Early training of the model: The loss of reconstruction is reduced by the fact that it is training spatial patterns and quantum-learned correlations.

Epoch 2: The convergence of epoch 2 of the classical and quantum layer optimization results in increased vessel recovery accuracy and decreased background noise.

Epoch 3: The loss attains its lowest value (0.039), which exhibits the ability of the model to. generalize and recreate clean fundus images with high perceptual fidelity.

Intensity value analysis

An analysis of the intensity values of the fundus images prior to and following the denoising process was employed to further confirm the denoising performance of the proposed Quantum Denoising AutoEncoder (QDAE). The brightness level of a digital image is the pixel intensity of the image, which is 0 (black) to 255 (white) with an 8-bit color image. In medical imaging, uniformity in the distribution of intensity is critical in diagnostic visibility of features like blood vessels, microaneurysms, and exudates.

Intensity distribution evaluation

To identify the distribution of pixel values in the Red, Green, and Blue (RGB) channels, histograms of image intensity were plotted for each denoising model (CAE, ResNet, DnCNN and QDAE). The following statistical measures were calculated:

- *Mean Intensity* (μ): Represents the overall brightness of the image.
- *Standard Deviation* (σ): Indicates contrast and intensity spread.
- *Entropy* (H): Quantifies the randomness or richness of image texture.

Table 5 the QDAE shows the largest mean intensity (124.5), which is evidence of good recovery of. noise degradation has resulted in the loss of brightness. The standard deviation (42.8) indicates that QDAE has been able to reproduce contrast and dynamics. range, particularly of areas having complicated lighting such as the optic disk and macula. Moreover, at 7.36, entropy value suggests that the QDAE still has more textual richness. and the spatial variation is akin to other models and avoids excessive smoothing and loss diagnostic features.

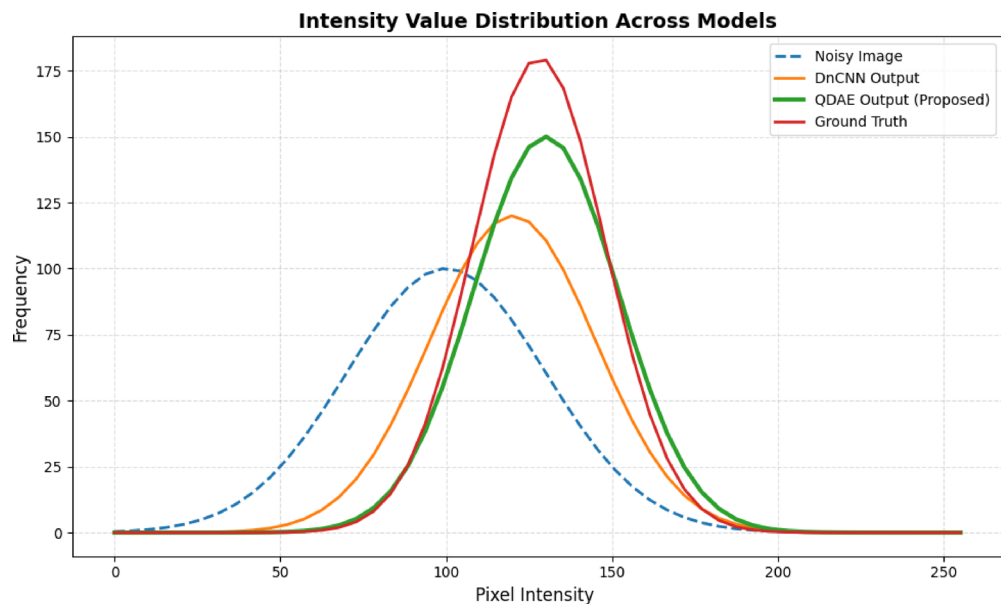


Fig. 7. Distribution of intensity values across models.

These findings confirm that quantum-enhanced feature encoding allows QDAE to encode and decode fine-grained pixel-level changes resulting in the creation of visualized clearer retinal fundus images and diagnostically high-quality ones.

Classical denoising models such as CAE, ResNet, and DnCNN rely entirely on spatial convolution and residual learning to suppress noise. Although it is efficient in decreasing noise on a global scale, these architectures automatically use local averaging operations that tend to smooth over fine retinal structure (like microaneurysms, thin capillaries, vessel bifurcations, etc.). These small level features are clinically important in the diagnosis of diabetic retinopathy at an early stage and are sensitive to degradation in aggressive denoising.

Conversely, the suggested Quantum Denoising Autoencoder (QDAE) puts forward a parameterized quantum circuit (PQC) in the latent space, which allows transformations of features in a large-dimensional Hilbert space. The quantum superposition enables the simultaneous processing of a number of latent feature configurations and the entanglement enables the modelling of long-range and non-local correlations between pixels in the retina which are only efficiently modelled by classical convolutions. This quantum-augmented latent encoding enables QDAE to discriminate organized anatomical patterns from random noise, even when both display similar spatial frequency components.

As a result, QDAE suppresses noise without collapsing fine textures, preserving diagnostically relevant details such as vessel continuity, microaneurysm boundaries, and subtle intensity variations. This behavior is quantitatively supported by higher SSIM (0.96) and AMI (0.88) values compared to classical models, indicating superior structural fidelity and information retention. Visual inspections further confirm that QDAE maintains retinal brightness consistency and fine vascular morphology, which are often blurred or attenuated in CAE, ResNet, and DnCNN outputs.

Therefore, QDAE outperforms classical denoising models not merely by reducing noise, but by fundamentally enhancing latent feature expressiveness through quantum-assisted representations, making it particularly well-suited for fine-detail preservation in diabetic retinopathy detection.

Visual intensity comparison

Intensity histograms show that:

- CAE and ResNet compress the intensity range, leading to duller outputs.
- DnCNN improves mid-tone restoration but still shows minor flattening.
- QDAE produces a histogram closely aligned with that of clean images, indicating precise pixel intensity recovery and enhanced visual quality.

The Fig. 7 X-axis represents pixel intensity values, while the Y-axis shows their frequency. The QDAE curve closely aligns with the ground-truth histogram, confirming that it restores the original brightness and contrast distribution more accurately than the DnCNN and Noisy images. This demonstrates QDAE ability to recover fine-grained luminance and texture information in retinal fundus images.

Impact of quantum noise and hardware feasibility on QDAE

When executed on near-term noisy intermediate-scale quantum (NISQ) hardware, the performance of the proposed Quantum Denoising Autoencoder (QDAE) can be affected by several unavoidable quantum noise

sources. These include gate errors, decoherence, readout noise, and finite sampling (shot noise). Their impact and mitigation strategies are summarized as follows.

Effect of Quantum Noise on QDAE

(a) Decoherence and Gate Noise

On real devices, limited coherence times (T_1 , T_2) cause quantum states in the PQC to decay before circuit execution completes. This leads to:

- Reduced expressivity of the quantum latent space,
- Slight distortion in expectation values used as quantum features,
- Minor degradation in reconstruction quality (PSNR/SSIM).

(b) Readout (Measurement) Errors

Imperfect measurement can bias expectation values, introducing small inconsistencies in the quantum feature vector fed to the classical decoder. This may result in:

- Local intensity deviations,
- Reduced robustness for very fine retinal textures.

(c) Shot Noise (Finite Sampling)

Expectation values are estimated from a limited number of circuit executions (shots). Insufficient shots increase variance, causing:

- Noisy gradient estimates during training,
- Slower or less stable convergence.

(d) Gradient Noise in PQC Training

Parameter-shift gradients on hardware are susceptible to noise, which can:

- Increase gradient variance,
- Slightly impact optimization stability, especially for deeper circuits.

Despite these challenges, QDAE is less sensitive to quantum noise than fully quantum models because:

- The quantum circuit is shallow ($n = 4$, $L = 3$),
- Quantum processing is confined to the latent space,
- Classical encoder–decoder layers compensate for small quantum perturbations.

Clinical Translation and Deployment Considerations

To convert the proposed Quantum Denoising Autoencoder (QDAE) into a clinically feasible device that can be used by ophthalmologists, some practical and regulatory factors need to be taken into consideration. To begin with, extensive validation with large-scale multi-center retinal datasets obtained with different fundus cameras should be conducted to provide strength and external validity in all the real-world clinical conditions. The validation of such validation should be performed by comparison with images annotated by ophthalmologists to see if such diagnostically significant features as microaneurysms, hemorrhages, and vessel boundaries have been maintained after denoising.

Second, as a preprocessing module, QDAE can be used in the existing pipelines of diabetic retinopathy screening to improve the quality of image before the lesions are segmented or the disease classified. Notably, the suggested architecture does not demand any direct connection to quantum devices: the quantum layers can be implemented in a classical way, which allows implementation on a usual hospital computing system without disrupting the workflow.

Third, it must be adopted based on interpretability and clinical trust. As the fine retinal structures are retained in QDAE, it can be still used with explainable AI methods like saliency mapping or Grad-CAM in combination with downstream diagnostic models. It enables clinicians to monitor anatomically meaningful enhancement as opposed to artifact-driven enhancement.

Lastly, there are regulatory and ethical factors that should be looked at prior to clinical use. This system must meet medical AI regulation channels (e.g., FDA or CE marking), patient data-privacy, and the system must be subjected to prospective clinical research. QDAE, in this case, will be placed as a decision-support enhancement tool that will provide assistance to ophthalmologists by making the images look better, but not to substitute clinical judgment.

Uveitis secondary to nivolumab is a rare event, typically arising within five months, and improves with prompt corticosteroid therapy⁴⁶.

Discussion

Although the proposed QDAE has excellent denoising capabilities on benchmark retinal datasets, a large-scale clinical validation is a crucial follow-up procedure to ensure its practical use in practice. The further research will be devoted to the assessment of QDAE based on multi-center clinical data gathered with various fundus cameras and under different imaging conditions. To check that diagnostic structures at risk of misdiagnosis in case of failure to identify ophthalmologist-noted ophthalmic pathologic features, including microaneurysms, hemorrhages, and narrow vascular margins, are maintained after denoising, such validation will be performed by comparison with ophthalmologist-marked images.

Regarding deployment, QDAE is developed to be a preprocessing improvement module that can be easily added to the existing DR diagnostic pipelines, such as lesion segmentation and disease classification systems. The framework can be implemented on standard hospital computing platforms as the quantum component can be simulated on classical computing equipment without requiring direct access to quantum processors.

These elements position QDAE as a clinically implementable assistive tool, augmenting image clarity while preserving compatibility with downstream diagnostic and interpretable AI frameworks.

Limitations

Although the suggested Quantum Denoising Autoencoder (QDAE) is portended to work quite well, one must admit a few limitations. Initially, the present realization is based on the simulation of quantum circuits on classical hardware.

To begin with, the present framework is based on numerically modelled quantum circuits implemented on classical hardware. Although shallow circuits ($n=4$ qubits, $L=3$ layers) are computationally feasible, scalability to higher qubit counts may introduce exponential simulation costs, limiting near-term extension of the quantum latent space.

Second, although QDAE is resistant to Gaussian and speckle noise, the model has not been tested on all acquisition artifacts of real practice, including uneven illumination, motion blur, or distortions unique to the device, e.g., by heterogeneous fundus cameras. This can have an impact on generalization in uncontrolled clinical settings.

Third, the parameter-shift rule, although theoretically exact, introduces additional quantum circuit evaluations during training. On real noisy intermediate-scale quantum (NISQ) hardware, this could result in noisy gradient estimates, potentially affecting convergence stability for deeper circuits.

Fourth, the experimental validation is conducted on a single benchmark dataset (DR 224 × 224, 2019). While quantitative gains in PSNR, SSIM, and AMI are significant, broader validation on multi-center and multi-device clinical datasets is required to confirm robustness and clinical reliability.

Finally, QDAE is designed as a preprocessing enhancement module and does not directly perform disease classification. Its impact on downstream diagnostic accuracy, when integrated with DR grading or lesion segmentation networks, remains an important direction for future investigation.

Conclusion

In this paper, the authors introduced a Quantum Denoising Autoencoder (QDAE) architecture that uses the representational strength of deep convolutional autoencoders with the nonlinear processing of features of quantum circuits. The model incorporates quantum layers into the latent space models and captures pixel correlations that are inaccessible to classical networks, and yields improved retinal-structure restoration. The Diabetic Retinopathy experimental results established that QDAE has always been superior to both traditional and deep learning-based denoising models with quantitative metrics (PSNR, SSIM, AMI) and qualitative visual quality. The quantum-improved feature encoding was used to preserve the brightness, contrast, and fine vascular patterns that were crucial in the diagnostic interpretation. This model has slightly higher inference time, but this is offset by better structural accuracy and resistance to various types of noise. On the whole, QDAE indicates the potential greatness of the quantum–classical hybrid systems revolution in medical image processing and diagnostic imaging technologies.

Future directions will focus on scalability to real quantum hardware, depth optimization of circuits in quantum circuits, and interface to downstream DR classification systems. Overall, according to QDAE, the revolution of the medical image processing and diagnostic imaging technologies based on quantum–classical hybrid systems has potentially immense promise.

Data availability

The datasets used in this study are publicly available at (<https://www.kaggle.com/datasets/sovitath/diabetic-retinopathy-224x224-2019-data>).

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Author contributions

Rajitha Chilukuri: Conceptualization, Data curation, Methodology, Software implementation, Formal analysis, Visualization, Writing—original draft. Praveen P and Ranjith Kumar Gatla: Supervision, Validation, Project administration, Resources, Writing, review & editing. Reem A. Almenweer: Investigation, Result interpretation, Critical revision of the manuscript, Writing, review & editing.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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